

Instance i

one task;
every arm runs on it, so
arms are
compared in
pairs

Arm A

1 call

one attempt

no checks, no retry

Arm B

up to k calls

attempt

write and run its
own checks

checks
pass?

yes, or k spent

no, calls left

records c_i ,
the calls B used

repair, with the
failures fed back

matched per instance: C gets exactly c_i calls on instance i

Arm C

exactly c_i calls

try 1

try 2

...

try c_i

self-consistency, else
the last attempt

unguided: no checks, no feedback between tries

Hidden-check scorer

checks the harness
never sees
per-instance score:
the fraction of hidden
checks passed
several runs per arm,
averaged within the
instance first

Pre-registered contrasts

primary

B – C

paired by instance: checking and repair
against the same number of calls spent on
unguided retries

secondary

B – A

the literature's design, confounded with
compute; a calibration check
also reported

C – A

the effect of compute alone
target effect +7 points

Pilot, arm A only: the baseline score against the pre-registered band

BCB-Hard-Instruct @ haiku $n = 23$
BCB-Hard @ haiku $n = 16$
BCB-Hard-Instruct @ sonnet $n = 23$
BCB-Hard @ sonnet $n = 16$
MBPP @ haiku $n = 25$
BCB-Instruct @ haiku $n = 25$
MBPP+ @ haiku $n = 25$
MBPP+ @ sonnet $n = 25$
MBPP @ sonnet $n = 25$

pre-registered band 25–70%

share of hidden checks passed 0 0.5 1

● inside the band on the point estimate: 1 of 9 cells
bars: 95% t intervals; wholly inside the band: 0 of 9
confirmatory comparison: not run, as the protocol requires when the pilot fails